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A clinical environment simulator for dynamic AI evaluation

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Clinical evaluation of large language models (LLMs) currently relies on static datasets and isolated scenarios that fail to capture the cascading effects of healthcare decisions. We propose the Clinical Environment Simulator (CES), a framework that evaluates clinical LLMs within digital hospital environments where every decision dynamically alters future states. The CES would use a parallel simulation architecture: a ‘hospital engine’ that tracks bed availability, staff workloads and equipment status in real time, and a ‘patient engine’ that simulates disease progression and treatment responses based on LLM interventions. Unlike current benchmarks, the CES framework requires clinical LLMs to execute decisions through realistic electronic health record interfaces, while managing trade-offs between individual patient optimization and system-wide efficiency. The CES enables three critical evaluations absent from current benchmarks: temporal reasoning under evolving constraints, where delayed diagnostics can lead to patient deterioration; resource-aware decision-making, where aggressive workups for one patient may exhaust capacity needed by others; and operational resilience, through adversarial testing with simultaneous emergencies and system failures. By scoring LLM performance on both clinical outcomes and operational metrics, the CES represents a shift toward evaluating clinical LLMs as a dynamic and integrated component of healthcare delivery systems.

Recent advances demonstrate that LLMs have achieved strong performance on diagnostic and clinical tasks[1–10], reporting performance exceeding that of individual clinicians[5,10] or demonstrating that AI-assisted workflows outperform clinicians working alone[4]. This is attributable to evaluation frameworks that support AI system development[11–18]. However, our primary methods for evaluating clinical LLMs rely on static, historical datasets—collections of disconnected ‘snapshots’ of care that are stripped of temporal context, interactivity and consequence[19–26].

In clinical practice, patient care decisions are not binary, immediate or static but rather represent cascading effects over time and across systems. A hospitalized patient’s morning lab results may reshape the entire trajectory of care, influencing afternoon monitoring protocols, evening medication adjustments and subsequent discharge planning. The decision to admit rather than discharge fills a bed that might be critically needed hours later. Individual patient trajectories—from initial presentation through diagnosis, treatment and discharge—collectively determine the system dynamics that enable or constrain

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future care delivery. These temporal and systemic interdependencies remain invisible to current LLM evaluation paradigms.

This disconnect between static evaluation and dynamic clinical reality parallels challenges overcome in other safety-critical domains. In the field of aviation, pilot training was transformed by replacing written examinations with comprehensive flight simulators that recreate the full complexity of flying, including dynamic weather systems, mechanical failures and evolving air traffic patterns[27,28]. These simulators evaluate not only knowledge but also the ability to maintain performance while managing multiple simultaneous challenges within resource constraints.

Medicine requires an analogous transformation for clinical LLMs. Clinical excellence aspires to navigate the continuous temporal evolution of patient care within resource constraints. Current benchmarks[11–18,24–26] cannot assess whether clinical LLMs understand how individual decisions aggregate into system-level effects, how resource constraints shape optimal care pathways, or how present interventions impact future capacity. This gap between isolated reasoning tasks and integrated clinical practice represents a critical barrier to meaningful LLM deployment in healthcare[29]. To bridge this gap, we propose a Clinical Environment Simulator (CES) framework to fundamentally reimagine how we evaluate clinical LLMs. We outline the key features of this framework and strategies for implementation. We compare the CES with existing evaluation frameworks and discuss potential limitations and opportunities for future development.

CES

Rather than testing performance on isolated scenarios, CES uses a persistent digital hospital environment where AI decisions continuously reshape both patient trajectories and system states, capturing the temporal dynamics that define clinical care (Fig. 1).

From isolated scenarios to integrated environments

Current LLM benchmarks[11–18] present clinical reasoning as a series of disconnected puzzles—static patient presentations with predetermined information and fixed outcomes. While these isolated scenarios effectively measure diagnostic accuracy, they fail to capture three essential aspects of clinical practice: temporal evolution, where patient states change based on intervention timing; system constraints, where resource availability shapes feasible options; and cascading effects, where individual decisions trigger downstream consequences across the healthcare system.

Our proposed CES framework transforms evaluation by embedding clinical LLMs within digital hospital environments that evolve continuously. Unlike traditional benchmarks that assume optimal diagnostic pathways and timely interventions, the CES simulates the reality that patients can deteriorate while awaiting gold-standard assessments. A patient presenting with chest pain might progress from stable angina to acute myocardial infarction during the wait for noninvasive cardiac testing, forcing immediate decisions to escalate therapies based on incomplete information. Similarly, a patient with suspected pulmonary embolism could decompensate while awaiting computed tomography angiography, necessitating empiric thrombolysis despite incomplete information. Every decision creates cascading effects: admissions fill beds, procedures occupy operating rooms and investigations strain laboratory and imaging capacity, dynamically reshaping the landscape of available interventions.

This temporal realism enables the CES to evaluate scenarios impossible with static datasets. These counterfactual scenarios—where the 'textbook' approach becomes untenable owing to evolving patient instability, fluctuating patient flow and dynamic resource constraints—reflect the pressures and trade-offs inherent in everyday clinical practice. By modeling these dynamic trajectories (see Fig. 2 for an example of dynamic patient trajectories), the CES evaluates not just diagnostic accuracy, but also adaptive clinical reasoning under time constraints and evolving uncertainty.

CES implements this dynamic evaluation through a parallel simulation architecture. The 'hospital engine' maintains real-time tracking of bed availability, staff workload, equipment status and departmental workflows—not as static values but as dynamic states influenced by admission patterns, discharge timing and procedural schedules. The 'patient engine' simulates individual clinical trajectories through physiological models that respond to interventions, delays and the natural progression of disease states.

When clinical LLMs act within the proposed CES, both engines would coordinate state transitions. A decision to admit a patient triggers bed assignment in the hospital engine while initiating monitoring protocols in the patient engine. Ordering urgent imaging consumes scanner time while potentially revealing findings that alter the patient's trajectory. This parallel simulation architecture creates authentic feedback loops where individual clinical decisions influence system capacity, which in turn constrains future options, mirroring the complex interdependencies that challenge real clinical practice.

From text generation to action execution

Contemporary clinical LLMs[1–10] generally operate in text space, generating diagnoses and treatment recommendations as natural language outputs. Although these capabilities demonstrate impressive medical reasoning, they do not address the practical requirements of clinical care delivery, where recommendations must be executed through electronic health record (EHR) interfaces.

In clinical practice, medical decisions require translation into interface interactions, navigating menus, selecting options, entering parameters and completing multistep workflows. Current LLM benchmarks evaluate text generation capabilities or multiple-choice question performance but do not assess whether systems can execute these recommendations within EHR environments. This represents a notable gap between LLM evaluation methodologies and deployment requirements.

CES addresses this gap by accommodating the heterogeneous reality of healthcare information technology through flexible action spaces that mirror actual clinical system interfaces. Clinical LLMs under evaluation can interact through application programming interfaces when available, executing structured commands for laboratory ordering, medication administration and clinical documentation workflows. Alternatively, these LLMs can operate through graphical user interface interaction when programmatic access is unavailable, accomplishing clinical actions through screen interpretation and interface manipulation. This creates a self-improving cycle where clinical LLMs can be trained against increasingly sophisticated versions of themselves within the simulation environment, with each iteration of the model serving as a more challenging opponent for the next. The simulation becomes both training ground and adversary, generating challenging clinical scenarios that push the boundaries of current capabilities. As the LLM masters standard protocols, the CES can dynamically introduce variations that force the development of more robust clinical reasoning strategies[30,31].

Balancing patient outcomes and operational efficiency in decision-making

Clinical care inherently involves trade-offs between optimizing current patient care and preserving system capacity[32], operating across immediate (pain management), short-term (complication prevention), medium-term (discharge planning) and long-term (resource scheduling) horizons. Emergency departments exemplify this ethical complexity, as high patient volumes force a shift from egalitarian approaches (optimizing each individual patient) toward utilitarian decision-making (greatest good for the greatest number)[33–36]. Diagnostic workups offer

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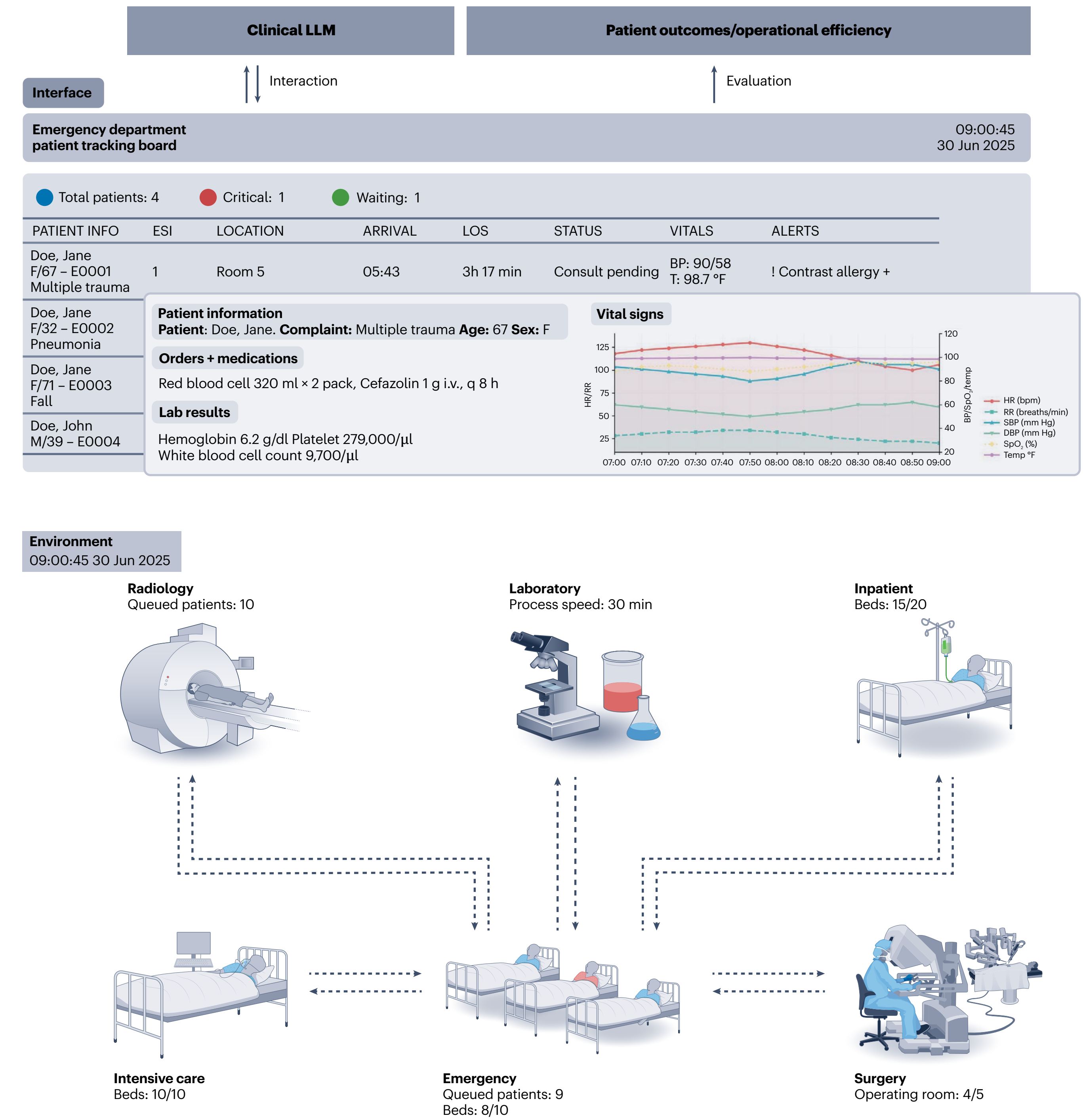

Fig. 1 | Paradigm of a CES. The proposed CES embeds algorithmic agents in a simulation of the hospital, enabling simultaneous assessment of patient outcomes (diagnostic accuracy, clinical trajectories, treatment effectiveness) and operational efficiency (resource utilization, patient flow, team workload) across time. BP, blood pressure; bpm, beats per min; DBP, diastolic blood pressure; ESI, emergency severity index; HR, heart rate; i.v., intravenous; LOS, length of stay; RR, respiratory rate; SBP, systolic blood pressure; SpO_2, peripheral capillary oxygen saturation; T, temperature.

clarity but strain lab capacity during surge conditions, and aggressive early interventions for one patient might exhaust resources needed by others. What functions well during standard conditions can paralyze operations during surge conditions.

We propose a CES that explicitly models these competing priorities through dual metrics that assess both patient outcomes and operational efficiency. Unlike traditional benchmarks that focus solely on diagnostic accuracy or treatment appropriateness, CES simultaneously measures how AI decisions impact individual patient health trajectories and system-wide resource utilization. This dual-metric approach forms the core of the CES evaluation framework. Success is measured through composite scoring derived from simulation event logs. Patient outcomes include survival (binary exit state), time to treatment (minutes from presentation to first disease-specific intervention) and guideline adherence (mandatory steps completed within specified windows). Operational metrics track length of stay (days from admission to discharge), emergency department throughput (patients per day), resource utilization (time-weighted occupancy of

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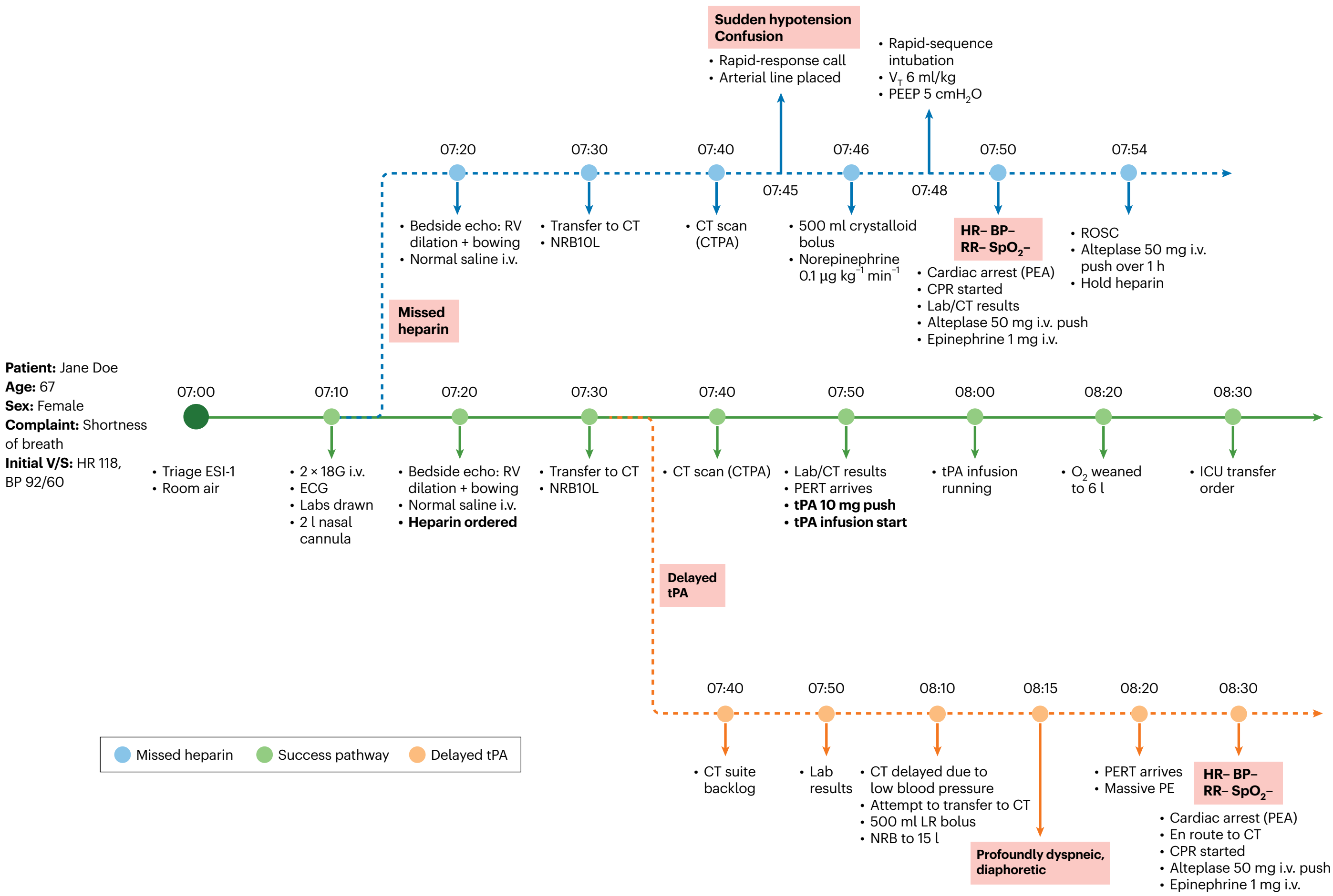

Fig. 2 | Dynamic patient trajectory demonstrating temporal decision-making within a CES. An AI agent must diagnose and treat massive pulmonary embolism in an overcrowded emergency department with limited resources. The timeline branches into three pathways based on treatment timing: green indicates timely intervention leading to stabilization; blue indicates missed initial treatment requiring emergency rescue; and orange indicates delayed treatment resulting in clinical deterioration. This scenario evaluates whether clinical LLMs can navigate resource constraints, anticipate clinical deterioration and prioritize time-critical interventions when optimal care pathways become unavailable. The initial layout was created in Canva (https://www.canva.com/design/DAG8_92XDiM/VXUD0jx3Fi-u-6JmokMWog/edit/). CPR, cardiopulmonary resuscitation; CT, computed tomography; CTPA, CT pulmonary angiography; ECG, electrocardiogram; ICU, intensive care unit; LR, lactated Ringer's solution; NRB, non-rebreather mask; PE, pulmonary embolism; PEA, pulseless electrical activity; PERT, pulmonary embolism response team; PEEP, positive end-expiratory pressure; ROSC, return of spontaneous circulation; RV, right ventricle; tPA, tissue plasminogen activator; V/S, vital signs; V_T, tidal volume.

beds and equipment) and episode cost (cumulative charges from presentation through discharge). The framework rewards decisions that improve patient care without compromising system function, while penalizing approaches that excel in one dimension at the expense of the other.

CES thus provides a foundation for developing clinical LLMs that demonstrate systematic intelligence rather than mere diagnostic accuracy, ensuring clinical AI readiness extends beyond standardized test performance to encompass the operational competence required for deployment within the temporal fabric of healthcare delivery.

Implementation and continuous improvement

The technical challenges of maintaining simulation fidelity, validating against real clinical data streams, and managing error propagation across coordinated engines are substantial but addressable through systematic engineering. The patient engine of the CES could combine predefined disease trajectory templates with partial grounding in existing EHR data and language model generation of plausible counterfactual scenarios. Expert-defined trajectory templates could encode evidence-based disease progression patterns through conditional branching based on intervention timing and appropriateness. Initial patient presentations, comorbidity patterns and baseline laboratory values can be drawn from actual clinical encounters to ensure realistic starting conditions. Language models then generate diverse counterfactual evolution paths within trajectory constraints, producing variations in symptom progression, treatment responses and complication patterns that reflect biological variability while maintaining clinical plausibility. Expert physician evaluation serves as a validation mechanism, assessing whether generated scenarios exhibit appropriate physiological behavior and realistic temporal dynamics.

The hospital engine implements discrete-event simulation informed by empirical operational data extracted from EHR timestamps. This engine maintains dynamic state variables for all shared resources including computed tomography scanners, laboratory processing capacity, operating rooms and inpatient beds. When clinical actions are submitted, the engine allocates resources that reflect actual workflow patterns, such as blood work triggering sequential allocation of nurses and lab technicians, with processing durations derived from real clinical encounters. The engine could incorporate priority-based allocation logic where critical cases for conditions like stroke or aortic dissection can preempt routine studies, creating realistic queues and delays that emerge naturally from competing demands.

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Table 1 | Comparison of the proposed CES with other healthcare AI environments

Features	Static benchmarks	Conversational diagnosis	Sequential diagnosis	CES
Clinical outcome metrics	✔	✔	✔	✔
Sequential information gathering	✘	✔	✔	✔
Multi-turn decision-making	✘	✔	✔	✔
Resource availability tracking	✘	✘	✘	✔
Operational efficiency metrics	✘	✘	✘	✔
Patient state evolution over time	✘	✘	✘	✔
Treatment response trajectories	✘	✘	✘	✔
Adversarial scenario stress testing	✘	✘	✘	✔

The two engines could be synchronized through a temporal orchestrator that advances simulation time in parallel, coordinating state updates in a sequential manner where each time point processes clinical actions, updates resource allocations, advances patient physiological states and generates observations for the clinical LLM. This orchestration architecture enables generation of customized scenarios spanning entire clinical shifts with multiple simultaneous patient encounters, where new patients arrive according to realistic patterns, existing patients evolve based on their disease trajectories and received interventions, and resource constraints dynamically reshape available options. The assessment would be then generated at the end of the simulation in clinical metrics such as diagnostic accuracy, treatment timeliness, length of stay of patients, and cost.

CES should also enable continuous improvement through mechanisms that validate and refine its clinical accuracy. The simulator can run temporal simulations with actual clinical cases captured in EHRs, comparing predicted state transitions against observed evolution when sufficient retrospective data exists. Systematic discrepancies between simulated and actual trajectories identify areas requiring refinement in trajectory templates, language model predictions or resource modeling. Additionally, expert clinicians provide feedback on the clinical plausibility of simulated patient responses, rating the realism of vital sign trajectories, symptom evolution patterns and treatment response dynamics to align simulated behavior more closely with clinical experience. These complementary approaches ensure that the simulator maintains fidelity to clinical practice as medical knowledge evolves and healthcare delivery systems adapt to changing demands.

Comparison with existing evaluation benchmarks

Clinical AI evaluation has progressed through increasingly sophisticated paradigms, each addressing specific limitations while introducing new capabilities (Table 1).

Static benchmarks[11,12] assess clinical outcome metrics through fixed datasets, providing controlled evaluation with clear endpoints. Conversational diagnosis frameworks[37,38] introduce sequential information gathering and multi-turn decision-making through patient–doctor dialogs. Sequential diagnosis approaches[14,16,17,20] further incorporate cost modeling, enabling evaluation of resource-conscious diagnostic strategies. The field has also recognized the importance of realistic EHR environments for clinical AI evaluation, with MedAgentBench[39] providing an interface based on fast healthcare interoperability resources (a standard framework for securely passing data between systems) that evaluates AI performance on discrete clinical operations such as patient information retrieval and order placement with static patient records.

The CES extends these frameworks by introducing three new dimensions. First, temporal dynamics: by modeling continuous patient state evolution, treatment response trajectories and time-dependent clinical deterioration, the CES requires AI systems to reason about intervention timing and disease progression. Second, resource modeling: by implementing real-time hospital capacity tracking across beds, equipment and staff, the CES necessitates resource-aware decision-making where individual patient care must be balanced against system-wide constraints. Third, adversarial stress testing: by introducing simultaneous emergencies and system failures, the CES demands AI robustness across the full spectrum of operational conditions from routine to crisis scenarios.

By introducing new environmental dimensions that necessitate expanded AI–environment interaction and multidimensional evaluation spanning clinical outcomes and operational efficiency metrics, the CES complements existing frameworks to create a more comprehensive assessment of clinical AI readiness for real-world deployment. Nevertheless, we also acknowledge what the proposed CES cannot do: unlike physics-based simulators that predict vehicle trajectories with mathematical precision, the CES cannot perfectly predict individual patient responses. Human physiology is irreducibly complex, with outcomes depending on factors that resist deterministic modeling. We present the CES not as an endpoint but as an achievable and valuable next step that advances healthcare AI evaluation beyond static benchmarks toward the dynamic, temporally aware assessment that healthcare deployment demands.

Opportunities

Clinical LLMs in the CES

The proposed CES opens new avenues for developing and evaluating the next generation of clinical LLMs. By providing interfaces where LLMs can learn to navigate, the CES could enable the development of AI agents that seamlessly integrate across the fragmented landscape of healthcare system. This capability presents opportunities to create clinical LLMs that adapt to any clinical environment, from cutting-edge academic centers with fully integrated systems to community hospitals relying on decades-old interfaces[40].

Beyond basic integration, the CES creates opportunities to develop clinical LLMs that truly understand the complexity of clinical workflows. Future AI agents could learn how single high-level clinical decisions explode into multitudes of low-level software interactions, while recognizing when seemingly distinct clinical decisions can be accomplished in one interface action, such as how 'discontinue all home medications' might address 20 orders at once. This deep understanding of workflow patterns positions LLMs to eventually optimize clinical interactions while informing the design of EHR systems that finally align with clinical thought processes.

Forging robustness through adversarial testing

A major advantage of the proposed CES framework is its ability to conduct adversarial testing of clinical LLMs under difficult conditions[41,42]. Just as robotics researchers push their robots to breaking point in simulation before real-world deployment, the CES could generate challenging clinical scenarios to probe an LLM's weaknesses. Real-world

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hospital data often skew toward common scenarios, leaving rare but critical situations under-examined. In the CES, one can increase the frequency of high-stakes edge cases on demand, such as acute anaphylactic shock with drug allergies or an older trauma patient on anticoagulants developing compartment syndrome. It can also simulate multiple simultaneous emergencies, evaluating whether the LLM can maintain performance in worst-case scenarios that human clinicians train for but rarely encounter[41].

CES could enable precise control over testing conditions through customizable simulation parameters—adjusting case complexity, modifying resource constraints such as staffing levels and configuring system loads from standard to surge conditions. Evaluators can also introduce environmental disruptions such as EHR outages or equipment failures to assess AI resilience. This flexibility creates a co-evolutionary dynamic where, as clinical LLMs improve, the CES systematically increases difficulty and introduces novel failure modes. Through this iterative adversarial testing process, the CES develops clinical LLMs that demonstrate robustness across the full spectrum of healthcare delivery conditions, from optimal to crisis scenarios, ensuring they maintain performance when real hospitals face their most challenging moments.

Building human–AI partnerships through simulation

CES creates a structured pathway for clinical translation of LLM-based medical AI by enabling preclinical experimentation before real-world deployment. As a risk-free testing environment, the CES allows developers to systematically evaluate the safety and efficacy of clinical LLMs across diverse scenarios, identify failure modes and edge cases that would be unethical or impractical to test with actual patients, and iteratively refine AI systems without patient risk. This preclinical validation extends beyond individual LLM performance to encompass system-level effects: institutions can simulate how AI integration might reshape clinical workflows, test different implementation strategies, assess impacts on resource allocation and patient throughput and identify necessary infrastructure modifications—reducing the risks associated with the substantial organizational investments required for AI adoption. By serving as a virtual test bed where both AI capabilities and healthcare system adaptations can be explored in parallel, the CES bridges the gap between algorithmic development and clinical deployment.

Beyond LLM development and validation, CES may enable the codesign of future health systems and the physicians who will work within them. By serving as a virtual healthcare replica, this framework could be used to optimize clinical workflows and train physicians[14,43,44] in systems-level reasoning, preparing them for an environment where AI agents handle routine tasks. This optimization could recenter human clinicians on tasks that truly require human empathy, judgment and presence, while the LLM handles mundane digital tasks. Today's physicians and nurses spend enormous time clicking checkboxes, entering orders, documenting notes and navigating billing or compliance forms—activities that, while necessary, erode face-to-face contact with patients[45]. In a future enabled by LLMs tested and refined within the CES, many of these administrative and documentation tasks could be reliably offloaded to AI agents. These agents, having shown in simulation that they can safely manage EHR workflows and information exchange, would act as digital intermediaries for clinicians[46]. The intended result is that caregivers can devote more attention to the uniquely human aspects of medicine: making eye contact, actively listening to patients' stories, providing physical comfort and conveying empathy—activities that have been shown to improve clinical outcomes but are increasingly marginalized by digital workflows.

Conclusion

The development of clinically viable clinical LLMs requires a fundamental shift from static validation paradigms to dynamic, temporally aware evaluation frameworks that capture the interconnected nature of healthcare delivery—where individual patient decisions create cascading effects. We propose the CES as a pathway toward this goal. Ultimately, the crucial measure of clinical LLM readiness lies in their ability to function as a reliable partner within the temporal dynamics of healthcare delivery.

References

1. Goh, E. et al. Large language model influence on diagnostic reasoning: a randomized clinical trial. *JAMA Netw. Open* **7**, e2440969 (2024).
2. McDuff, D. et al. Towards accurate differential diagnosis with large language models. *Nature* **642**, 451–457 (2025).
3. Cabral, S. et al. Clinical reasoning of a generative artificial intelligence model compared with physicians. *JAMA Intern. Med.* **184**, 581–583 (2024).
4. Goh, E. et al. GPT-4 assistance for improvement of physician performance on patient care tasks: a randomized controlled trial. *Nat. Med.* **31**, 1233–1238 (2025).
5. Tu, T. et al. Towards conversational diagnostic artificial intelligence. *Nature* **642**, 442–450 (2025).
6. Gao, S. et al. TxAgent: an AI agent for therapeutic reasoning across a universe of tools. Preprint at https://doi.org/10.48550/arXiv.2503.10970 (2025).
7. Singhal, K. et al. Large language models encode clinical knowledge. *Nature* **620**, 172–180 (2023).
8. Sandmann, S. et al. Benchmark evaluation of DeepSeek large language models in clinical decision-making. *Nat. Med.* **31**, 2546–2549 (2025).
9. Tordjman, M. et al. Comparative benchmarking of the DeepSeek large language model on medical tasks and clinical reasoning. *Nat. Med.* **31**, 2550–2555 (2025).
10. Liu, X. et al. A generalist medical language model for disease diagnosis assistance. *Nat. Med.* **31**, 932–942 (2025).
11. Pal, A., Umapathi, L. K. & Sankarasubbu, M. MedMCQA: a large-scale multi-subject multi-choice dataset for medical domain question answering. In *Conference on Health, Inference, and Learning* 248–260 (PMLR, 2022).
12. Jin, Q., Dhingra, B., Liu, Z., Cohen, W. & Lu, X. PubMedQA: a dataset for biomedical research question answering. In *Proc. 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing* (eds Inui, K. et al.) 2567–2577 (ACL, 2019).
13. Jin, D. et al. What disease does this patient have? A large-scale open domain question answering dataset from medical exams. *Appl. Sci.* **11**, 6421 (2021).
14. Schmidgall, S. et al. AgentClinic: a multimodal agent benchmark to evaluate AI in simulated clinical environments. Preprint at https://doi.org/10.48550/arXiv.2405.07960 (2024).
15. Hager, P. et al. Evaluation and mitigation of the limitations of large language models in clinical decision-making. *Nat. Med.* **30**, 2613–2622 (2024).
16. Fan, Z. et al. AI Hospital: benchmarking large language models in a multi-agent medical interaction simulator. In *Proc. 31st International Conference on Computational Linguistics* 10183–10213 (ACL, 2025).
17. Li, J. et al. Agent Hospital: a simulacrum of hospital with evolvable medical agents. Preprint at https://doi.org/10.48550/arXiv.2405.02957 (2024).
18. Bedi, S. et al. Holistic evaluation of large language models for medical tasks with MedHELM. *Nat. Med.* https://doi.org/10.1038/s41591-025-04151-2 (2026).

19. Zhang, S. et al. Rethinking human-AI collaboration in complex medical decision making: a case study in sepsis diagnosis. In *Proc. 2024 CHI Conference on Human Factors in Computing Systems* 445, 1–18 (ACM, 2024).
20. Nori, H. et al. Sequential diagnosis with language models. Preprint at https://doi.org/10.48550/arXiv.2506.22405 (2025).
21. Bedi, S., Mlauzi, I., Shin, D., Koyejo, S. & Shah, N. H. The optimization paradox in clinical AI multi-agent systems. Preprint at https://doi.org/10.48550/arXiv.2506.06574 (2025).
22. Rosenthal, J. T., Beecy, A. & Sabuncu, M. R. Rethinking clinical trials for medical AI with dynamic deployments of adaptive systems. *NPJ Digit. Med.* **8**, 252 (2025).
23. Palepu, A. et al. Towards conversational AI for disease management. Preprint at https://doi.org/10.48550/arXiv.2503.06074 (2025).
24. Johnson, A. E. W. et al. MIMIC-III, a freely accessible critical care database. *Sci. Data* **3**, 160035 (2016).
25. Johnson, A. E. W. et al. MIMIC-IV, a freely accessible electronic health record dataset. *Sci. Data* **10**, 1 (2023).
26. Kansal, A., Chen, E., Jin, B. T., Rajpurkar, P. & Kim, D. A. MC-MED, multimodal clinical monitoring in the emergency department. *Sci. Data* **12**, 1094 (2025).
27. Lazic, D. A., Grujic, V. & Tanaskovic, M. The role of flight simulation in flight training of pilots for crisis management. *SFJD* **3**, 3624–3636 (2022).
28. Allerton, D. J. The impact of flight simulation in aerospace. *Aeronaut. J.* **114**, 747–756 (2010).
29. Mahmood, F. A benchmarking crisis in biomedical machine learning. *Nat. Med.* **31**, 1060 (2025).
30. Silver, D. et al. Mastering the game of Go without human knowledge. *Nature* **550**, 354–359 (2017).
31. Silver, D. et al. Mastering the game of Go with deep neural networks and tree search. *Nature* **529**, 484–489 (2016).
32. Page, B., Irving, D., Amalberti, R. & Vincent, C. Health services under pressure: a scoping review and development of a taxonomy of adaptive strategies. *BMJ Qual. Saf.* **33**, 738–747 (2024).
33. Morley, C., Unwin, M., Peterson, G. M., Stankovich, J. & Kinsman, L. Emergency department crowding: a systematic review of causes, consequences and solutions. *PLoS ONE* **13**, e0203316 (2018).
34. Pines, J. M. et al. The impact of emergency department crowding measures on time to antibiotics for patients with community-acquired pneumonia. *Ann. Emerg. Med.* **50**, 510–516 (2007).
35. Bernstein, S. L. et al. The effect of emergency department crowding on clinically oriented outcomes. *Acad. Emerg. Med.* **16**, 1–10 (2009).
36. Emanuel, E. J. et al. Fair allocation of scarce medical resources in the time of Covid-19. *N. Engl. J. Med.* **382**, 2049–2055 (2020).
37. Johri, S. et al. An evaluation framework for clinical use of large language models in patient interaction tasks. *Nat. Med.* **31**, 77–86 (2025).
38. Arora, R. K. et al. HealthBench: Evaluating large language models towards improved human health. Preprint at https://doi.org/10.48550/arXiv.2505.08775 (2025).
39. Jiang, Y. et al. MedAgentBench: a virtual EHR environment to benchmark medical LLM agents. *NEJM AI* **2**, 9 (2025).
40. Zhang, C. et al. API agents vs. GUI agents: divergence and convergence. In *ICML 2025 Workshop on Computer Use Agents* (ICML, 2025).
41. Finlayson, S. G. et al. Adversarial attacks on medical machine learning. *Science* **363**, 1287–1289 (2019).
42. Javed, H., El-Sappagh, S. & Abuhmed, T. Robustness in deep learning models for medical diagnostics: security and adversarial challenges towards robust AI applications. *Artif. Intell. Rev*. **58**, 12 (2024).
43. Kumar, A. et al. OrderRex clinical user testing: a randomized trial of recommender system decision support on simulated cases. *J. Am. Med. Inform. Assoc.* **27**, 1850–1859 (2020).
44. Elendu, C. et al. The impact of simulation-based training in medical education: a review. *Medicine* **103**, e38813 (2024).
45. Sinsky, C. et al. Allocation of physician time in ambulatory practice: a time and motion study in 4 specialties. *Ann. Intern. Med.* **165**, 753–760 (2016).
46. Tierney, A. A. et al. Ambient artificial intelligence scribes: Learnings after 1 year and over 2.5 million uses. *NEJM Catal. Innov. Care Deliv*. https://doi.org/10.1056/CAT.25.0040 (2025).

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